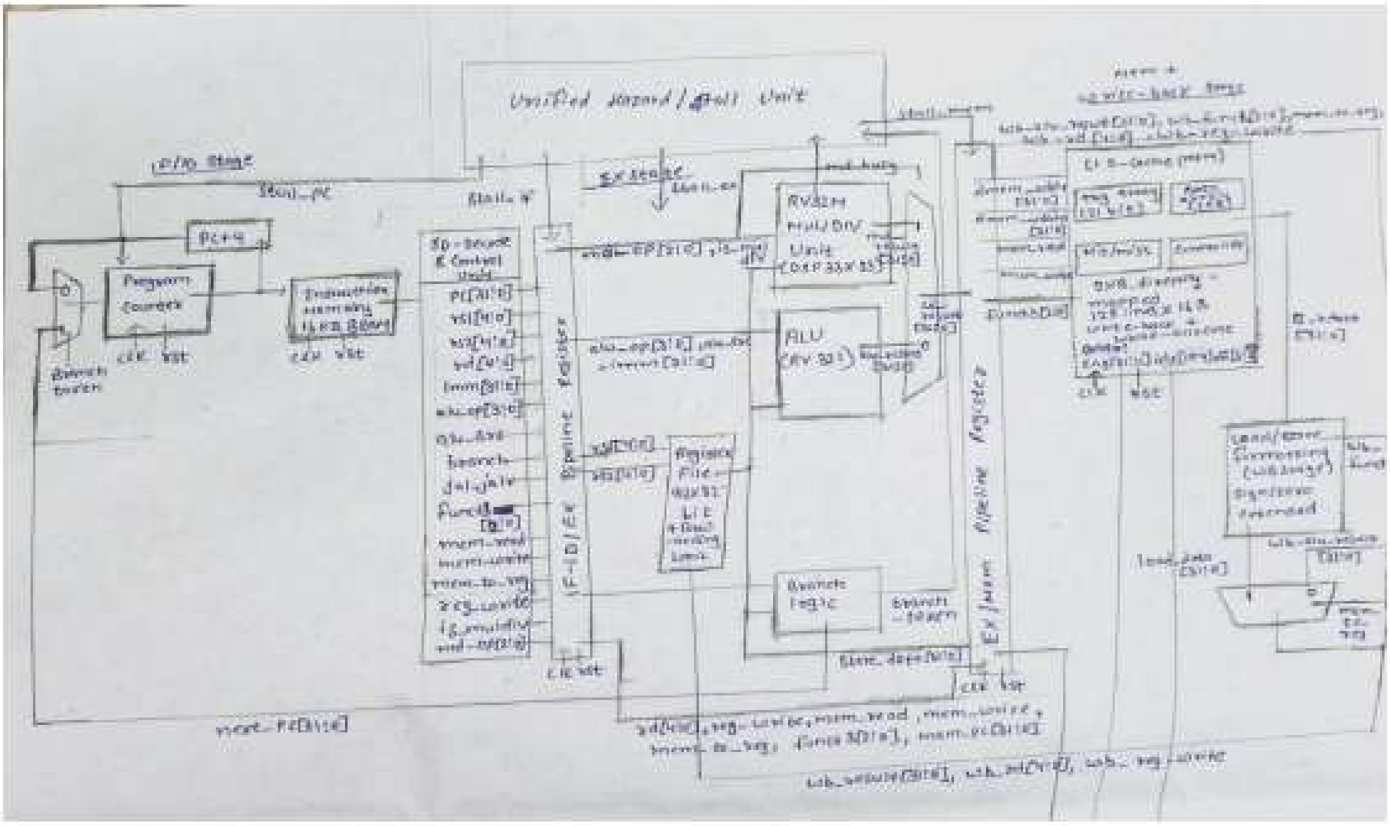
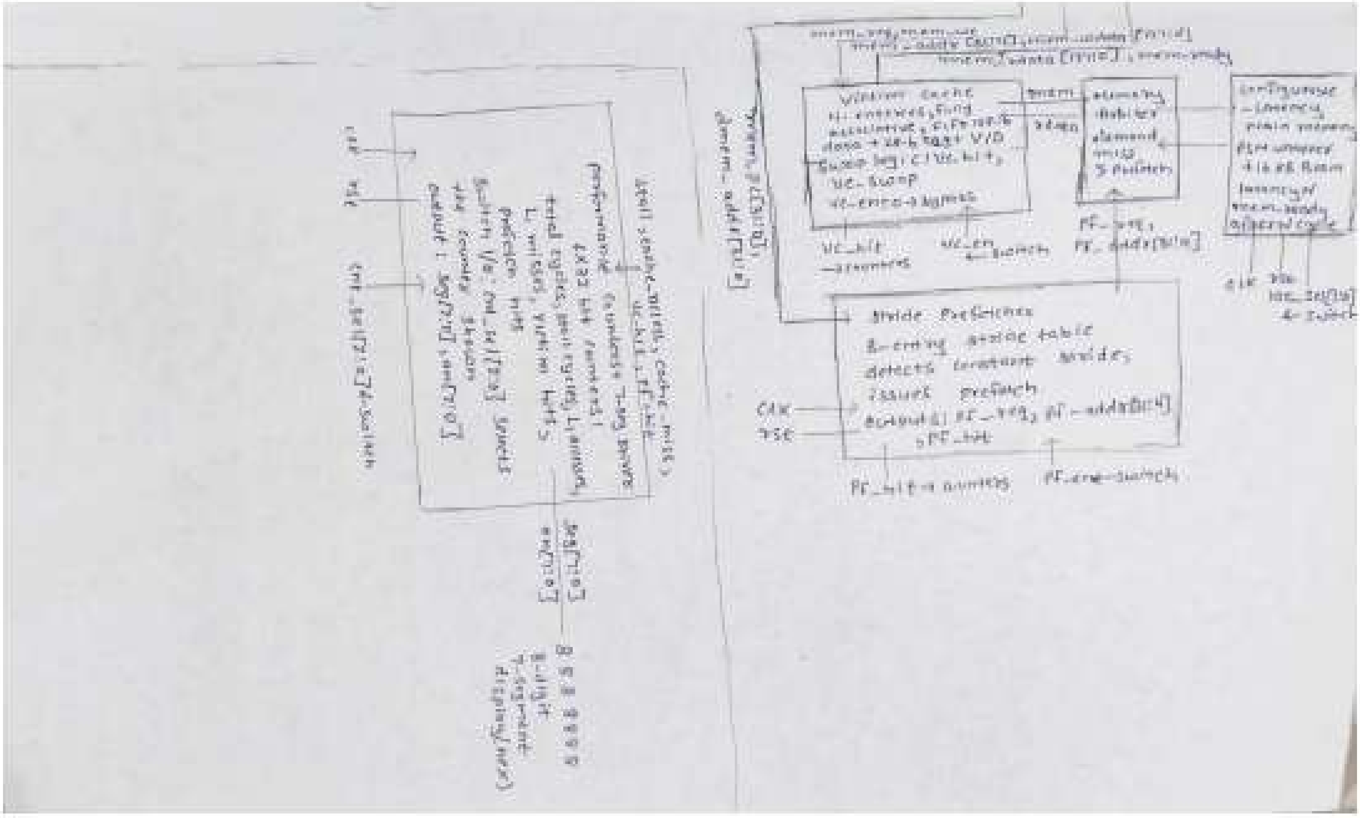
1. Top-Level Hardware Block Diagram (FULL PAGE)

Diagram must be clean, readable, and properly labeled

Must clearly show:

- Pipeline stages (IF, ID, EX, MEM, WB)
- All major modules (ALU, MUL/DIV, Control, Memory)
- Signal names + widths
- Pipeline registers (IF/ID, ID/EX, etc.)
- Control signals (at least key ones: RegWrite, MemRead, etc.)
- Clock and reset
- **New modules highlighted clearly**


Unified Hazard / Stall Unit
IF/ID Stage
stall_pc
stall_if
EX Stage
stall_ex
PC+4
Program Counter
Branch taken
clk rst
Instruction Memory
clk rst
ID - Decode & Control Unit
pc[31:0]
rs1[4:0]
rs2[4:0]
rd[4:0]
imm[31:0]
alu_op[3:0]
branch
jal_jalr
funct3
mem_read
mem_write
mem_to_reg
reg_write
is_muldiv
IF/ID/EX Pipeline Register
clk rst
alu_op[3:0]
rs1[4:0]
rs2[4:0]
Register File 32x32
RV32M MUL/DIV Unit
md_busy
ALU (RV 32I)
Branch logic
branch_taken
EX/MEM Pipeline Register
clk rst
mem + write-back stage
L1 D-Cache
DDR memory - mapped
clk rst
load/store processing
sign/zero extended
load_data [31:0]
next_pc[31:0]
wb_result[31:0], wb_rd[4:0], wb_reg_write


Victim cache
4-entries, fully
associative, FIFO repl
data + 26-b tags V/D
Swap logic (VC_hit,
VC_swap
VC_hit
→ counters
VC_en
← switch
memory
arbiter
demand
miss
> prefetch
PF_req,
PF_addr[31:4]
Configurable
latency
main memory
CLK
RST
Stride Prefetcher
8-entry stride table
detects constant stride;
issues prefetch
outputs: PF_req, PF_addr[31:4]
, PF_hit
CLK
RST
PF_hit → counters
PF_en ← switch
8-digit
7-segment
display (hex)